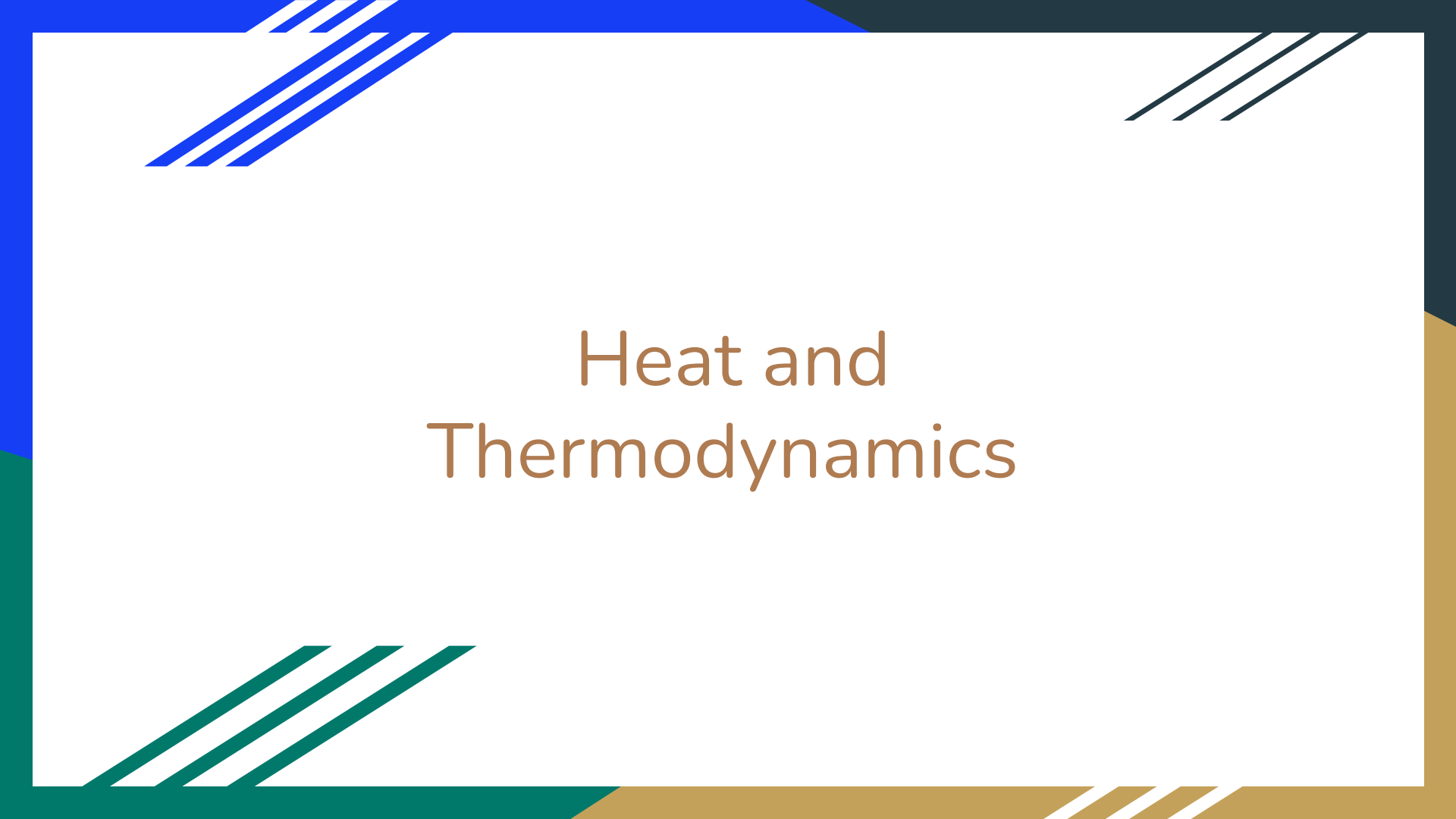




# Heat and Thermodynamics

# Introduction

- Heat and thermodynamics are branches of physics that deal with the study of energy transfer and transformation.
- In this presentation, we will explore key concepts including the First Law of Thermodynamics, internal energy, specific heats of gases, and work done by expanding gas.

# First Law of Thermodynamics

- The First Law of Thermodynamics states that energy cannot be created or destroyed, only transformed from one form to another.
- It is often expressed as  $\Delta U = Q - W$ , where  $\Delta U$  is the change in internal energy of a system,  $Q$  is the heat added to the system, and  $W$  is the work done by the system.
- This law forms the basis for understanding energy conservation in various physical processes.

# Units of Heat

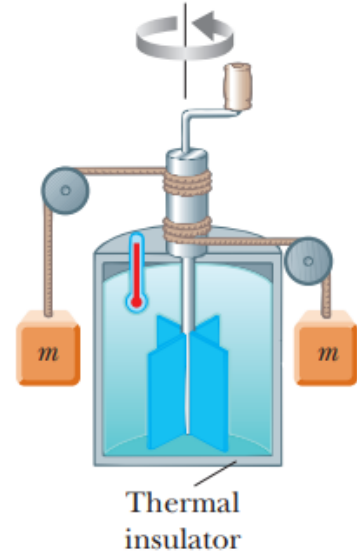
- Calorie
- The amount of energy transfer necessary to raise the temperature of 1 g of water from  $14.5^{\circ}\text{C}$  to  $15.5^{\circ}\text{C}$
- the amount of energy transfer required to raise the temperature of 1 lb of water from  $63^{\circ}\text{F}$  to  $64^{\circ}\text{F}$

# Joule

It is equal to the amount of work done when a force of one newton displaces a mass through a distance of one metre in the direction of that force.

# The Mechanical Equivalent of Heat

Joule found that the loss in mechanical energy is proportional to the product of the mass of the water and the increase in water temperature. The proportionality constant was found to be approximately  $4.18 \text{ J/g} \cdot ^\circ\text{C}$



# Internal Energy

- Internal energy ( $U$ ) of a system is the sum of its kinetic and potential energies at the microscopic level.
- It is a state function, meaning it depends only on the current state of the system, not on how the system reached that state.
- Changes in internal energy can occur through heat transfer ( $Q$ ) or work done ( $W$ ) on or by the system, as per the First Law of Thermodynamics.

# Heat capacity

The heat capacity  $C$  of a particular sample is defined as the amount of energy needed to raise the temperature of that sample by  $1^{\circ}\text{C}$ .

The specific heat  $c$  of a substance is the heat capacity per unit mass. Therefore, if energy  $Q$  transfers to a sample of a substance with mass  $m$  and the temperature of the sample changes by  $T$ , the specific heat of the substance is  $Q = C \Delta T$



## Specific Heats of Some Substances at 25°C and Atmospheric Pressure

| Substance               | Specific Heat $c$ |            | Substance           | Specific Heat $c$ |            |
|-------------------------|-------------------|------------|---------------------|-------------------|------------|
|                         | J/kg · °C         | cal/g · °C |                     | J/kg · °C         | cal/g · °C |
| <i>Elemental solids</i> |                   |            | <i>Other solids</i> |                   |            |
| Aluminum                | 900               | 0.215      | Brass               | 380               | 0.092      |
| Beryllium               | 1 830             | 0.436      | Glass               | 837               | 0.200      |
| Cadmium                 | 230               | 0.055      | Ice (−5°C)          | 2 090             | 0.50       |
| Copper                  | 387               | 0.092 4    | Marble              | 860               | 0.21       |
| Germanium               | 322               | 0.077      | Wood                | 1 700             | 0.41       |
| Gold                    | 129               | 0.030 8    | <i>Liquids</i>      |                   |            |
| Iron                    | 448               | 0.107      | Alcohol (ethyl)     | 2 400             | 0.58       |
| Lead                    | 128               | 0.030 5    | Mercury             | 140               | 0.033      |
| Silicon                 | 703               | 0.168      | Water (15°C)        | 4 186             | 1.00       |
| Silver                  | 234               | 0.056      | <i>Gas</i>          |                   |            |
|                         |                   |            | Steam (100°C)       | 2 010             | 0.48       |

# Specific heat

The specific heat  $c$  of a substance is the heat capacity per unit mass. Therefore, if energy  $Q$  transfers to a sample of a substance with mass  $m$  and the temperature of the sample changes by  $T$ , the specific heat of the substance is

$$C \equiv Q/m\Delta T$$